

BYT2

Physics: Mechanics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A delivery drone built by a startup with 30 employees accelerates straight up from rest at 2.0 m/s^2 for 5.0 s while carrying a 1.2-kg package. How far does it rise in that time?

- A. 50 m
- B. 25 m
- C. 10 m
- D. 5 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A car brakes from 20 m/s to a stop in 40 m with constant deceleration. What is the size of the deceleration?

- A. 2.0 m/s^2
- B. 10 m/s^2
- C. 0.5 m/s^2
- D. 5.0 m/s^2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A warehouse worker wearing new steel-toe boots pushes a 50-kg crate across the floor with a 200-N horizontal force. Kinetic friction on the crate is 150 N. What is the crate's acceleration?

- A. 1.0 m/s^2
- B. 7.0 m/s^2
- C. Zero, because friction always cancels an applied push.
- D. 4.0 m/s^2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

At an amusement park that draws a million visitors a year, a 500-kg roller-coaster car starts from rest at the top of a 20-m hill. Ignoring friction, how fast is it moving at the bottom? ($g = 9.8 \text{ m/s}^2$)

- A. About 14 m/s
- B. It depends on the 500-kg mass; a heavier car goes faster.
- C. About 19.8 m/s
- D. 392 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

In a rail yard that operates 24 hours a day, a 20,000-kg freight car rolling at 3.0 m/s couples with a stationary 10,000-kg car. How fast do they move together?

- A. 3.0 m/s
- B. 1.5 m/s
- C. 6.0 m/s
- D. 2.0 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A 2-kg ball moving east at 6 m/s bounces off a wall and moves west at 4 m/s. What is the size of the impulse on the ball?

- A. 8 kg·m/s
- B. 12 kg·m/s
- C. 4 kg·m/s
- D. 20 kg·m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A satellite run by a company that has launched 60 satellites weighs 800 N on Earth's surface. It now orbits at 2 Earth radii from Earth's center. What gravitational force does Earth exert on it there?

- A. 0 N, because objects in orbit have no gravity acting on them.
- B. 200 N
- C. 3,200 N
- D. 400 N

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

How far from Earth's center, in Earth radii, would an object weigh 1/9 of its surface weight?

- A. 3
- B. 1/3
- C. 9
- D. 81

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

On a playground seesaw renovated last year, a 30-kg child sits 2.0 m from the pivot. Where must a 40-kg child sit on the other side to balance it?

- A. 1.5 m from the pivot
- B. It cannot balance, since the children never weigh the same.
- C. 2.0 m from the pivot
- D. About 2.7 m from the pivot

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A lake-cabin owner floats a block of wood bought at a lumberyard 20 miles away. The block has volume 0.020 m^3 and density 600 kg/m^3 ; the lake water is $1,000 \text{ kg/m}^3$. What fraction of the block is under water?

- A. 2%
- B. 40%
- C. 60%
- D. 100%, since buoyant force always equals the weight of the block's full volume of water.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A clockmaker whose shop has been in the family for three generations has a 1.0-m pendulum with a 2-kg bob, swinging with a period of about 2.0 s. She swaps in a 4-kg bob of the same size. What happens to the period?

- A.** It increases by a factor of $\sqrt{2}$.
- B.** It stays about 2.0 s.
- C.** It halves, since heavier bobs swing faster.
- D.** It doubles, since the bob is twice as heavy.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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12. [TRANSFER — reversed stem / different variable]

By what factor must a pendulum's length change to double its period?

- A.** $\sqrt{2}$ times as long
- B.** $1/4$ as long
- C.** 4 times as long
- D.** 2 times as long

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A delivery drone ~~built by a startup with 30 employees~~ accelerates straight up from rest at 2.0 m/s^2 for 5.0 s ~~while carrying a 1.2 kg package~~. How far does it rise in that time?

Target word: distance from rest **Decides:** $d = \frac{1}{2} \times 2.0 \times 5.0^2$.

A	Dropped element	Drops the $\frac{1}{2}$.
B	Correct	$\frac{1}{2} \times 2 \times 25 = 25 \text{ m}$.
C	Adjacent category	That is the final speed in m/s , not the distance.
D	Confusion set	Uses t instead of t^2 .

2. Answer D Concept 1 · transfer

A car brakes from 20 m/s to a stop in 40 m with constant deceleration. What is the size of the deceleration?

Target word: solve for acceleration **Decides:** $0 = 20^2 - 2a(40)$.

A	Wrong stem number	Divides distance by speed.
B	Dropped element	Drops the factor of 2.
C	Inverted operation	Divides speed by distance.
D	Correct	$400 / 80 = 5.0$.

3. Answer A Concept 2

A warehouse worker ~~wearing new steel-toe boots~~ pushes a 50-kg crate across the floor with a 200-N horizontal force. Kinetic friction on the crate is 150 N . What is the crate's acceleration?

Target word: acceleration **Decides:** Net force = $200 - 150 = 50 \text{ N}$.

A	Correct	$50 \text{ N} / 50 \text{ kg}$.
B	Inverted operation	Adds friction to the push.
C	Absolute language	Friction only cancels when it equals the push.
D	Dropped element	Ignores friction.

4. Answer C Concept 3

At an amusement park ~~that draws a million visitors a year~~, a 500-kg roller-coaster car starts from rest at the top of a 20-m hill. Ignoring friction, how fast is it moving at the bottom? ($g = 9.8 \text{ m/s}^2$)

Target word: speed at bottom **Decides:** $v = \sqrt{2 \times 9.8 \times 20}$.

A	Confusion set	Drops the 2: $\sqrt{9.8 \times 20}$.
B	Camouflage	Mass cancels in $mgh = \frac{1}{2}mv^2$.
C	Correct	$\sqrt{392} \approx 19.8$.
D	Dropped element	Forgets the square root.

5. Answer D Concept 4

In a rail yard ~~that operates 24 hours a day~~, a 20,000-kg freight car rolling at 3.0 m/s couples with a stationary 10,000-kg car. How fast do they move together?

Target word: final speed **Decides:** 60,000 kg·m/s shared by 30,000 kg.

A	Dropped element	Ignores the added mass.
B	Confusion set	Averages the two speeds.
C	Inverted operation	Divides by the smaller mass alone.
D	Correct	Momentum is conserved: 60,000 / 30,000.

6. Answer D Concept 4 · transfer

A 2-kg ball moving east at 6 m/s bounces off a wall and moves west at 4 m/s. What is the size of the impulse on the ball?

Target word: impulse with a direction reversal **Decides:** $\Delta p = 2[(-4) - 6]$.

A	Confusion set	Uses only the final momentum.
B	Dropped element	Uses only the initial momentum.
C	Inverted operation	Subtracts speeds, ignoring the reversal.
D	Correct	$ 2 \times (-10) = 20$.

7. Answer B Concept 5

A satellite ~~run by a company that has launched 60 satellites~~ weighs 800 N on Earth's surface. It now orbits at 2 Earth radii from Earth's center. What gravitational force does Earth exert on it there?

Target word: force at doubled distance **Decides:** Force scales as $1/r^2$: 800 / 4.

A	Absolute language	Gravity provides the orbit; "weightless" means free fall.
B	Correct	Doubling r cuts force to one-quarter.
C	Inverted operation	Multiplies by 4 instead of dividing.
D	Dropped element	Uses $1/r$ instead of $1/r^2$.

8. Answer A Concept 5 · transfer

How far from Earth's center, in Earth radii, would an object weigh 1/9 of its surface weight?

Target word: solve for distance **Decides:** $1/r^2 = 1/9$.

A	Correct	$r^2 = 9$, so $r = 3$.
B	Inverted operation	Closer in would increase weight.
C	Dropped element	Forgets the square.
D	Confusion set	Squares instead of taking the root.

9. Answer A Concept 6

On a playground seesaw ~~renovated last year~~, a 30-kg child sits 2.0 m from the pivot. Where must a 40-kg child sit on the other side to balance it?

Target word: balance distance **Decides:** $30 \times 2.0 = 40 \times r$.

A	Correct	$60 / 40 = 1.5$.
B	Absolute language	Unequal weights balance at unequal distances.
C	Dropped element	Drops the different masses.
D	Inverted operation	Inverts the ratio: $40 \times 2 / 30$.

10. Answer C Concept 7

A lake-cabin owner floats a block of wood ~~bought at a lumberyard 20 miles away~~. The block has volume 0.020 m^3 and density 600 kg/m^3 ; the lake water is $1,000 \text{ kg/m}^3$. What fraction of the block is under water?

Target word: fraction submerged **Decides:** $600 / 1,000$.

A	Wrong stem number	Reads the 0.020 m^3 volume as a fraction.
B	Confusion set	That is the fraction above water.
C	Correct	A floater displaces its own weight: density ratio 0.60.
D	Absolute language	That holds only when fully submerged.

11. Answer B Concept 8

A clockmaker ~~whose shop has been in the family for three generations~~ has a 1.0-m pendulum with a 2-kg bob, swinging with a period of about 2.0 s. She swaps in a 4-kg bob of the same size. What happens to the period?

Target word: effect of bob mass **Decides:** $T = 2\pi\sqrt{L/g}$ has no mass term.

A	Adjacent category	Applies the mass-spring rule, $T = 2\pi\sqrt{m/k}$.
B	Correct	Length and g set the period; mass does not.
C	Inverted operation	Heavier bobs fall at the same rate.
D	Confusion set	Mass does not appear in the pendulum formula.

12. Answer C Concept 8 · transfer

By what factor must a pendulum's length change to double its period?

Target word: solve for length factor **Decides:** $T \propto \sqrt{L}$.

A	Confusion set	Takes the root of the factor instead of squaring it.
B	Inverted operation	Shorter length shortens the period.
C	Correct	$\sqrt{4} = 2$.
D	Dropped element	Forgets the square root.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	10	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	2	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Constant-acceleration kinematics	From rest, $d = \frac{1}{2}at^2$; in general $v^2 = v_0^2 + 2ad$.	Dropping the $\frac{1}{2}$ or confusing final speed with distance.	Acceleration and a time or distance, starting or ending at rest.
2	Newton's second law with friction	$a = F_{\text{net}} / m$, where F_{net} is the vector sum of all forces, including friction.	Dividing the applied force alone by the mass.	An applied force and a friction force on one object.
3	Conservation of mechanical energy	Without friction, $mgh = \frac{1}{2}mv^2$, so $v = \sqrt{2gh}$, independent of mass.	Thinking heavier objects reach the bottom faster.	A drop height and "ignore friction."
4	Momentum and impulse	Total momentum is conserved in collisions; impulse = change in momentum, with direction.	Averaging speeds, or ignoring a reversal of direction.	Objects that stick together, or an object that bounces back.
5	Inverse-square gravitation	Gravitational force varies as $1/r^2$; double the distance from the center, one-quarter the force.	Using $1/r$, or thinking orbiting objects feel no gravity.	A distance given in planet radii.
6	Torque and static equilibrium	In rotational equilibrium, clockwise torques equal counterclockwise torques: $F_1r_1 = F_2r_2$.	Inverting the ratio of weights and distances.	A seesaw, lever, or beam with loads at distances.
7	Buoyancy	A floating object displaces its own weight of fluid, so fraction submerged = object density / fluid density.	Thinking the buoyant force always equals the weight of the object's full volume of fluid.	An object of known density floating in a fluid.
8	Simple pendulum period	$T = 2\pi\sqrt{L/g}$; period depends on length, not on bob mass or small amplitude.	Applying the mass-spring rule to a pendulum.	A change in bob mass or pendulum length.

Course scope (public description): Physics: Mechanics introduces foundational concepts of mechanics, including motion, gravitation, work and energy, momentum and collisions, rotational motion, static equilibrium, fluids, and oscillation.